

4 Genes have a role in determining the overall phenotype of an organism.

(a) Using haemophilia as an example, explain the relationship between a gene, a protein and a phenotype.

..... [5]

(b) In the pea plant, *Pisum sativum*, the genotype **lele** results in plants with short stems (dwarf plants).

Explain how the **lele** genotype and its gene product results in dwarf plants.

..... [3]



(c) Dwarf plants also occur if the genes controlling growth are not activated (switched on).

Explain how DELLA proteins act as repressors preventing gene expression.

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..... [3]

[Total: 11]

